Week 2: Customer Churn

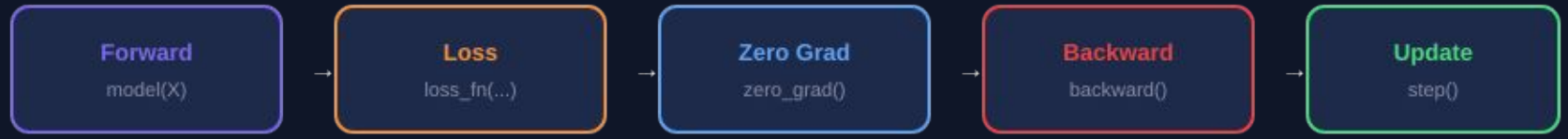
Same Training Loop, New Problem

IIT Madras · Wadhwani School of AI

Live Coding Session — Week 2

Quick Refresher: The 5-Step Loop

This is the only pattern you need to memorize



Same 5 steps. Week 1, week 2, always.

Problem	Loss Function	Output
Multi-class (Week 1)	`nn.CrossEntropyLoss()`	N logits ⟶ argmax
Binary (Today!)	**`nn.BCEWithLogitsLoss()`**	1 logit ⟶ sigmoid ⟶ threshold

What Changes, What Stays

Different data, different loss — same 5-step loop

Week 1: Fashion-MNIST

Task	**Multi-class (10 classes)**
Input	28×28 grayscale images
Features	784 pixels (raw)
Loss	CrossEntropyLoss
Output	10 logits → argmax
Data prep	Minimal (ToTensor + Normalize)
Balance	Balanced classes ✓

→

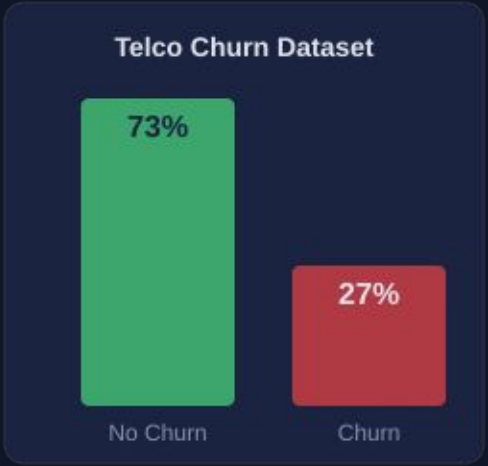
Week 2: Customer Churn

Task	**Binary (churn / no churn)**
Input	Tabular customer data
Features	Mixed: categorical + numeric
Loss	BCEWithLogitsLoss
Output	1 logit → sigmoid → threshold
Data prep	Encoding + scaling needed
Balance	73/27 imbalanced ✗

The training loop doesn't change. Same nn.Module, same optimizer, same 5 steps. Only the loss and data prep are different.

New Challenge: Class Imbalance

Fashion-MNIST was balanced. Real-world data almost never is.



The Problem

A model that always predicts "No Churn" gets 73% accuracy.

Sounds good — but it catches zero churners. Useless.

Accuracy is a lie with imbalanced data.

Our Solutions Today

1. Weighted loss — penalize missing a churner more heavily
2. Confusion matrix — see TP, FP, FN, TN, not just accuracy
3. Precision / Recall — metrics that actually matter for the business

```
BCEWithLogitsLoss(pos_weight=...)
```

Tells the model: "Missing a churner costs 2.7× more than a false alarm." Computed as (num_negative / num_positive).

Today's Coding Session

~10 min

Data Prep

Load Telco dataset, encode categoricals, scale features, handle imbalance

~15 min

Build & Train

nn.Module with BatchNorm + Dropout, BCEWithLogitsLoss, same 5-step loop

~12 min

Evaluate

Confusion matrix, precision/recall, why accuracy misleads on imbalanced data

~5 min

Week 1 vs 2

Side-by-side comparison: what changed, what stayed the same

+ 10 min Q&A at the end

Questions?

Week 2: Customer Churn Prediction

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